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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2025

CV1003-1 – ENGINEERING MECHANICS
 (For BT, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. _____ is said to be made up of infinite number of molecules packed in such a way that, there is no gap between the molecules.

A) Particle
 B) Continuum
 C) Point Force
 D) Rigid Body

2. If a number of forces are acting simultaneously at a point, it

A) Can be replaced by a single force
 B) Can be replaced by a couple
 C) Can be replaced by a couple and force
 D) Cannot be replaced

3. The resultant force is equal to the _____ of all the forces.

A) Product
 B) Division
 C) Sum
 D) Subtraction

4. The effect of a force on a body depends upon its _____

A) Position
 B) Direction
 C) Magnitude
 D) All the above

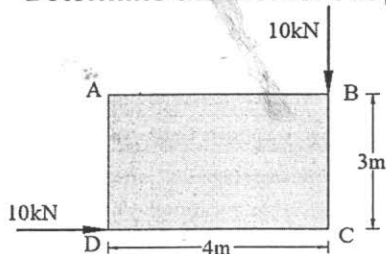
5. Moment of a force depends on _____

A) Moment center only
 B) Moment center and lever arm
 C) Lever arm only
 D) None of the above

6. An equivalent force couple system at a given point consists of _____

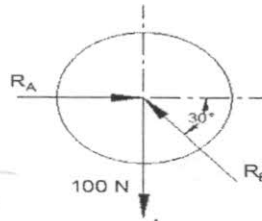
A) Same force-couple system
 B) Only Couples
 C) Same force with a moment
 D) Only forces

7. Determine the Moment at point C for the figure shown below.

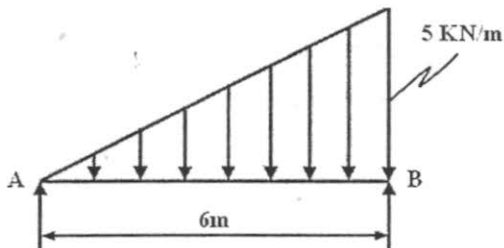


A) 0 kN-m
 B) 70 kN-m
 C) -30 kN-m
 D) -10 kN-m

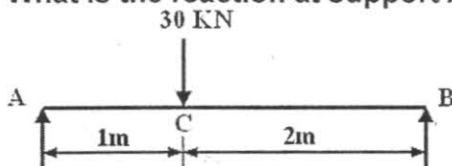
8. To keep the body in equilibrium, the magnitude of reaction B is _____



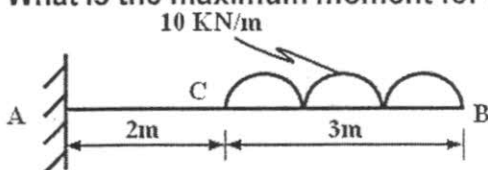
- A) 50 N
B) 15.47 N
C) 100 N
D) 115.47 N
9. Roller support is same as ____
A) Hinged Support
B) Simple support
C) Roller Support
D) Fixed Support
10. What is the vertical reaction at support A in the beam shown below?



- A) 10 kN
B) 15 kN
C) 50 kN
D) 60 kN
11. What is the reaction at support A in the beam shown below?



- A) 10 kN
B) 20 kN
C) 30 kN
D) 25 kN
12. Angle of friction is angle between ____
A) The weight of the body and the friction here
B) The incline and horizontal
C) Normal reaction and the resultant
D) The normal reaction and friction force
13. A cubical block rests on an inclined plane of $\mu = 1/\sqrt{3}$, determine the angle of inclination when the block just slides down the inclined plane
A) 30N
B) 40 N
C) 20N
D) 50 N
14. A force of 200 N is exerted on a snack box of 5 kg still on the floor. If the coefficient of friction is 0.3, calculate the static friction.
A) 30 N
B) 10 N
C) 20 N
D) 60 N
15. What is the maximum moment for the beam shown below?



- A) - 30 KN-m
B) - 60 KN-m
C) - 125 KN-m
D) - 105 KN-m

16. On simplification of the loading system, the net force acts at the ____ of the loading body.
- A) The mid-point
B) The central axis
C) Inertial point
D) Centroid
17. What is the vertical centroidal distance from base of right angled triangle of base 20 cm and side 40 cm?
- A) 19.36 cm
B) 13.33 cm
C) 13.28 cm
D) 38.72 cm
18. What is the moment of inertia of a semi-circular section of radius R about its centroidal axis parallel to diameter?
- A) $0.055 R^4$
B) $\pi R^4/16$
C) $\pi R^4/8$
D) $0.11 R^4$
19. What will be the moment of inertia of the rectangle with $b = 10$ mm and $d = 5$ mm about a horizontal axis passing through the base?
- A) 104.17 mm^4
B) 1782 mm^4
C) 416.67 mm^4
D) 1666.67 mm^4
20. Polar moment of inertia is calculation of moment of inertial about
- A) Z- axis
B) Y-axis
C) X-axis
D) Parallel axis

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Explain any four-force system with neat sketches.
b) List and explain the basic idealization in Engineering mechanics.
c) Determine the weight W_1 and W_2 such that string BC is horizontal shown in fig. 1(c), Assume the pulley to be frictionless and mass less.

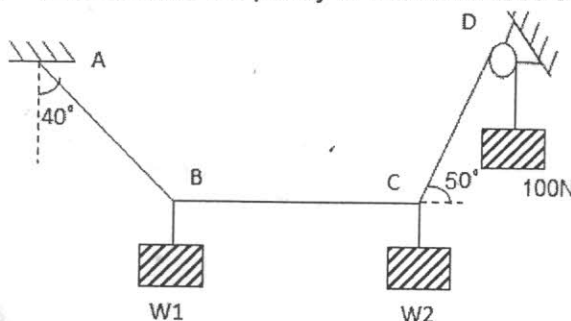


Fig. 1(c)

Marks	BT*	CO*	PO*
4	L*2	1	1,2
4	L2	1	1,2

2. a) Define equilibrium and mentioned the conditions of equilibrium.
b) Explain equivalent force-couple system with the help of neat sketches.
c) Determine the magnitude and direction of the resultant for the force system as shown in fig. Q 2 (c).

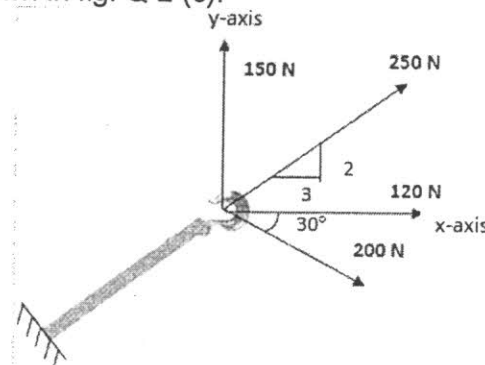


Fig. Q 2 (c).

8 L3 2 1,2

4 L1 2 1,2

4 L2 2 1,2

8 L3 1 1,2

3. a) State and explain principle of transmissibility of a force with neat sketch. State its limitations.
 b) Sketch the resultant of force system acting on the plate element with respect to point 'A' shown in fig. Q 3 (b).

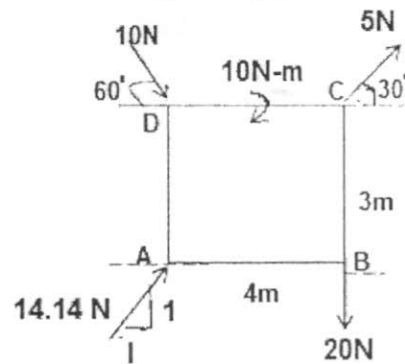


Fig. Q 3 (b).

Unit – II

4. a) Distinguish determinate and indeterminate beams with examples.
 b) Explain any two types of beam with neat sketch?
 c) A ladder 5 meters long rests on a horizontal ground and leans against a smooth vertical wall at an angle 70° with the horizontal. The weight of the ladder is 900 N and acts at its middle. The ladder is at the point of sliding, when a man weighing 750N stands on a rung 1.5 m from the bottom of the ladder. Calculate the coefficient of friction between the ladder and the floor.
5. a) State Coulomb's laws of friction (any four).
 b) Define i) coefficient of friction and ii) angle of friction
 c) Determine the resultant reactions at the supports for the loaded beam shown in fig. Q 5 (c).

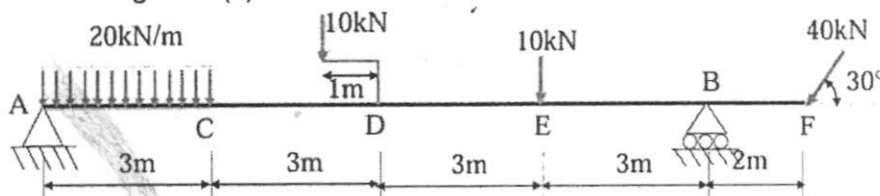


Fig. Q 5 (c)

6. a) Find the least magnitude of P required to cause the system of blocks shown in fig. Q 6 (a) to have impending motion to the right. The coefficient of friction under each block is 0.20.

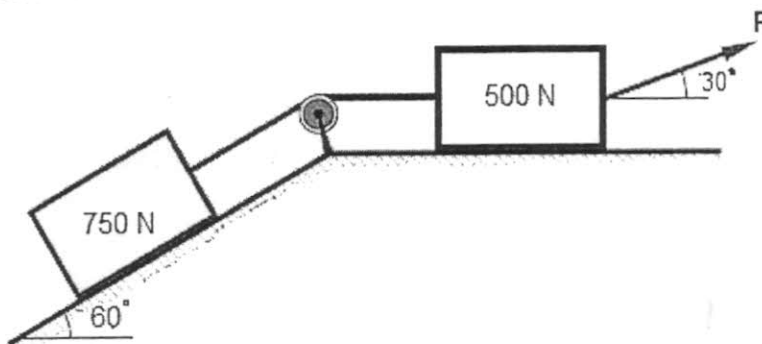


Fig. Q 6 (a)

- b) Determine the resultant reactions at the supports for the loaded beam shown in fig. Q 6 (b).

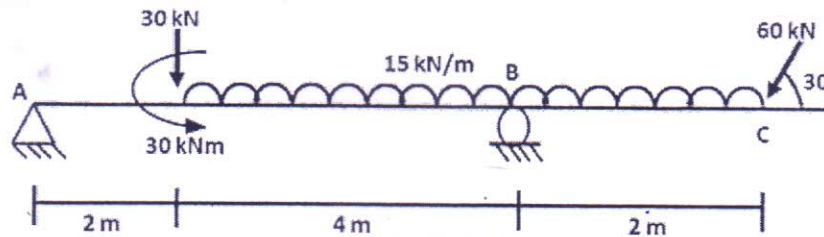


Fig. Q 6 (b)

8 L3 4 1,2

Unit – III

7. a) Develop an expression for moment of inertia of a semicircular lamina about its base from the method of integration.
b) Locate the centroid of the shaded area about the axes shown in fig. Q 7 (b).

8 L3 5 1,2

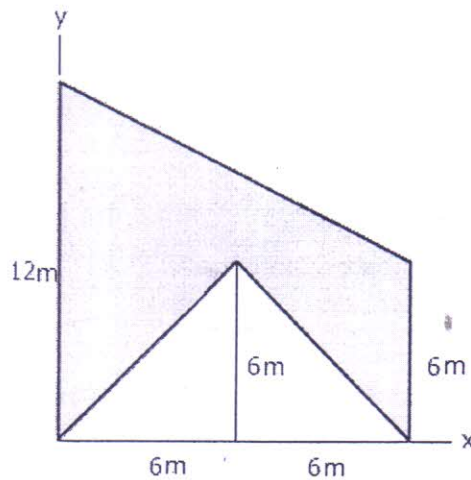


Fig. Q 7 (b)

8 L3 5 1,2

8. a) Prove that the centroid of a triangle of height h is at a distance $h/3$ from its base.
b) Determine the moment of inertia of given section in fig. Q 8 (b) with respect to horizontal centroidal axis.

8 L3 5 1,2

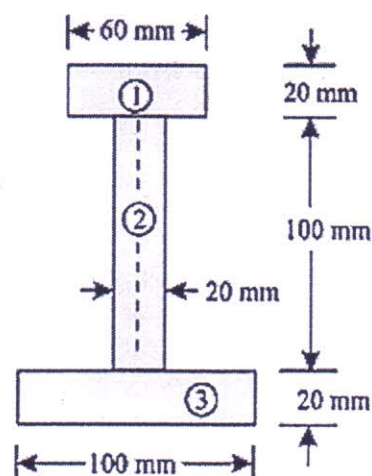


Fig. Q 8 (b)

8 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
